Vedant Singh Chauhan

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EDUCATION

Bachelor of Technology (Computer Science Engineering) **Nov '22 - May '26**
Indian Institute of Information Technology (IIIT), Vadodara 8.97

Minor Equivalent (Artificial Intelligence) **Nov '23 - May '26**
Indian Institute of Technology (IIT), Madras (NPTEL)

WORK EXPERIENCE

Engineering Intern **Jan '26 - Jun '26**
CRED Bangalore

- Built the Mosaic framework, a server-driven UI platform with typed widget builders, DAG-based data-source orchestration, design tokens, route targeting, and Studio authoring, reducing duplicate shared-data fetches from N calls to 1 and enabling screen changes without client releases.
- Engineered Propeller, a Go/gRPC real-time push platform with Redis/NATS pub-sub and native Flutter clients, reducing reconnect stalls from ~22s to ~2s and consolidating user/topic fan-out onto 1 stream.
- Engineered US Stocks workflows for INR-to-USD funding, Glomo checkout, Alpaca buy/sell orders, PPS polling, live portfolio updates, market data handling, and RBI LRS compliance up to $250K/year.

LLM Research Intern **Nov '25 - Apr '26**
Indian Institute of Information Technology (IIIT), Vadodara Gandhinagar

- Designed a 12M-parameter encoder-decoder Transformer to generate natural language activity descriptions from WiFi CSI signals, achieving a BLEU-4 of 0.67 (~51% over baseline).
- Engineered multi-scale temporal convolutions, Rotary Position Embeddings (RoPE), and stochastic depth on the UT-HAR dataset (4,973 samples, 7 activity classes), reaching 96.2% activity accuracy.
- Implemented curriculum learning and CSI-aware augmentation, raising BLEU-4 by +0.08 over baseline.

Mobile SDE Intern **May '25 - Oct '25**
Merlin AI (Google-backed) Bangalore

- Built image generation flows in Flutter, powering Bonkers; optimized generation pipelines, reducing latency by 26%; contributed to an in-house image optimization service, boosting adoption by 18%.
- Built core modules for Thine (thine.com) in SwiftUI, a privacy-first AI co-pilot; engineered its task router for seamless switching between chat, image gen, and doc tools, boosting efficiency and retention.
- Engineered immersive native widgets using Jetpack Glance (Android) and WidgetKit (iOS) via method channels, overcoming Flutter's limitations and increasing daily active widget interactions by 2.3x.

PROJECTS

Juno – Vehicle Pooling App | Flutter, Firebase, Socket.IO, Razorpay, Google Maps API **[Link]**

- Built and deployed a real-time ride-pooling app with live tracking, fare splitting, and payments.
- Implemented route optimization with Google Maps Directions, reducing average ride-match wait time.
- Published on Google Play Store, improving shared mobility adoption across student communities.

HopeLang – Programming Language & Web IDE | TypeScript, React, Monaco Editor, LangChain **[Link]**

- Designed a custom language with a hand-written lexer, recursive-descent parser, and tree-walking interpreter; built a pluggable module system for turtle graphics, music, and web generation.
- Built a browser-based IDE with Monaco Editor, real-time execution, AI coding assistant, and interactive game modes; published as an npm package, adopted by 1,700+ students across 9+ institutions.

TECHNICAL SKILLS

Mobile | Flutter, Dart, SwiftUI
Agents & Automation | n8n, LLM Agents, Workflow Automation, Tool Calling, MCP
AI | Transformers, TensorFlow, LangChain, LangGraph, RAG, LoRA/PEFT
Frontend | TypeScript, React.js, Three.js
Backend & Systems | Node.js, gRPC, Redis, NATS, WebSockets, Pub-Sub, ESP32
Databases | MongoDB, MySQL, Pinecone, Firebase
Data Structures & Algorithms | Java